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**DESIGN AND IMPLEMENTATION OF A PRIVACY-  
PRESERVING  
STRING-MATCHING FRAMEWORK USING SEARCHABLE  
ENCRYPTION  
AND TRUSTED EXECUTION ENVIRONMENTS (TEE)**

**A CAPSTONE PROJECT REPORT**

**Course: CSA5116 – Cryptography and Network Security**

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## **BONAFIDE CERTIFICATE**

This is to certify that the Capstone Project entitled “**DESIGN AND IMPLEMENTATION OF A PRIVACY-PRESERVING STRING-MATCHING FRAMEWORK USING SEARCHABLE ENCRYPTION AND TRUSTED EXECUTION ENVIRONMENTS (TEE)**” has been carried out by **Atchaya Chandran R (192521394)** under the supervision of Dr E Anbalagan and is submitted in partial fulfilment of the requirements for the course CSA5116-Cryptography and Network Security at SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai.

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## **DECLARATION**

I **Atchaya Chandran R (192521394)** the undersigned students of the Department of Electronics and Communication Engineering, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “Design and Implementation of a Privacy-Preserving String-Matching Framework Using Searchable Encryption and Trusted Execution Environments (TEE)” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged.

We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place: Chennai

Date: 04/0902026

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## INDIVIDUAL CONTRIBUTION STATEMENT

Mandatory for all team projects. The exact responsibilities and percentages should be finalized from the team's actual work records; no contribution percentages are fabricated in this draft.

| Student Name / Register No.    | Specific Responsibilities                                        | Design & Development Contribution         | Testing & Analysis Contribution              | Report Contribution                              | Approx. Contribution (%) |
|--------------------------------|------------------------------------------------------------------|-------------------------------------------|----------------------------------------------|--------------------------------------------------|--------------------------|
| Atchaya Chandran – R-192521394 | System architecture, secure-search integration and documentation | Architecture and module integration       | Functional search and security verification  | Chapters 1–5, figures and documentation          | 30%                      |
| Surabhi Surya P S – 192421267  | Searchable-encryption workflow and secure data/index handling    | Encryption and searchable-index module    | Encryption/search functional testing         | Literature review and implementation sections    | 25%                      |
| Vikram – 192411166             | TEE-based secure string-matching module                          | TEE and matching-engine implementation    | Matching accuracy and TEE operation testing  | Design, testing and results sections             | 25%                      |
| Susidharan V – 192521387       | Result verification, auditing and attack detection               | Verification, audit and monitoring module | Security, audit and attack-detection testing | Risk, security, conclusion and appendix sections | 20%                      |
| TOTAL                          |                                                                  |                                           |                                              |                                                  | 100%                     |

## **ABSTRACT**

The increasing use of cloud computing for storing and processing sensitive information has created a significant privacy challenge during data search and retrieval. Conventional string-matching approaches generally require sensitive information to be available in plaintext during processing, increasing the possibility of unauthorized disclosure. This is particularly important for applications involving healthcare records, financial documents, cloud-stored organizational information and other privacy-sensitive datasets. The project addresses this engineering problem by developing a privacy-preserving string-matching framework that enables secure search operations without directly exposing the underlying sensitive data.

The proposed framework combines Searchable Encryption (SE) with a Trusted Execution Environment (TEE) to provide confidentiality during both data storage and search processing. Sensitive information is encrypted before cloud storage, while a searchable index is generated to support efficient retrieval. Authorized users authenticate with the system and perform encrypted searches. The matching operation is performed within a protected TEE environment so that sensitive search processing is isolated from the untrusted cloud environment. The framework additionally incorporates secure authentication, cryptographic key management, encrypted indexing, result verification, audit logging and attack detection.

The system is organized into four principal modules: secure user authentication and cryptographic key management; secure data encryption and searchable index generation; privacy-preserving secure string matching; and secure result verification, auditing and attack detection. The implementation evidence presented in the project demonstrates a working secure keyword-search interface in which the TEE is active and matching results are returned to the authorized user.

The resulting framework demonstrates that searchable encrypted data can be queried while maintaining confidentiality and incorporating additional protected execution and security-monitoring mechanisms. The approach is intended for applications such as cloud storage, healthcare, financial information, genomic search, document retrieval and encrypted cloud analytics.

**Keywords:** Searchable Encryption, Trusted Execution Environment, Privacy-Preserving Search, Secure String Matching, Cloud Security, Cryptography

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## LIST OF ABBREVIATIONS / SYMBOLS

| Abbreviation | Description                                         |
|--------------|-----------------------------------------------------|
| SE           | Searchable Encryption                               |
| TEE          | Trusted Execution Environment                       |
| AES          | Advanced Encryption Standard                        |
| RSA          | Rivest–Shamir–Adleman                               |
| SHA          | Secure Hash Algorithm                               |
| ECC          | Elliptic Curve Cryptography                         |
| E2E          | End-to-End                                          |
| HIPAA        | Health Insurance Portability and Accountability Act |
| GDPR         | General Data Protection Regulation                  |
| NIST         | National Institute of Standards and Technology      |
| CSF          | Cybersecurity Framework                             |
| SGX          | Software Guard Extensions                           |

## **CHAPTER 1: INTRODUCTION AND ENGINEERING PROBLEM**

### **1.1 Background**

Cloud computing has become an important platform for storing and processing large quantities of information. However, storing sensitive information on third-party cloud infrastructure creates privacy and security concerns because the infrastructure hosting the data may not be fully trusted. Searching sensitive information using conventional techniques can require the data to be decrypted or exposed during processing, increasing the possibility of unauthorized disclosure.

Privacy-preserving string matching addresses this problem by allowing search operations to be performed without exposing sensitive plaintext information. Searchable Encryption (SE) provides mechanisms through which encrypted data can be searched without requiring the stored dataset to be directly decrypted. The project further introduces a Trusted Execution Environment (TEE), which provides an isolated and hardware-protected execution environment for sensitive operations.

The proposed framework integrates these technologies to provide a secure search architecture. Sensitive data is encrypted before being stored in the cloud, a searchable index is generated, and authorized queries are processed through the protected execution environment. The framework also incorporates authentication, key management, result verification, audit logging and attack detection. These components work together to provide confidentiality, controlled access and trustworthy retrieval.

### **1.2 Need for the Project**

- Privacy of sensitive information: sensitive information should not be exposed simply because a search operation needs to be performed.
- Untrusted cloud infrastructure: sensitive information stored in third-party cloud environments should remain protected even when the storage environment is not completely trusted.
- Secure search processing: encryption protects stored information, but useful information must still be extracted from encrypted datasets.
- Protection against unauthorized access: only authenticated and authorized users should perform search operations and retrieve results.
- Integrity and accountability: search results should be verified and user operations should be recorded through audit mechanisms.

### **1.3 Problem Statement**

Sensitive data stored in cloud environments is vulnerable to unauthorized access during search and processing. Traditional string-matching techniques generally require sensitive information to be available in plaintext during processing, thereby increasing the possibility of privacy breaches.

Existing searchable-encryption techniques can introduce performance overhead and may reveal limited information through search-related operations. Furthermore, without a protected execution environment, search processing may remain exposed to attacks against the computing environment.



Therefore, there is a need for a secure, privacy-preserving and efficient string-matching framework capable of searching encrypted data while minimizing plaintext exposure, protecting search processing and ensuring trustworthy result retrieval.

#### **1.4 Project Aim**

To design and implement a privacy-preserving string-matching framework that combines Searchable Encryption and Trusted Execution Environments to enable secure and accurate search over sensitive encrypted data while minimizing information exposure.

#### **1.5 Project Objectives**

15. To design a privacy-preserving framework for secure string matching on encrypted data.
16. To implement Searchable Encryption for searching sensitive information without directly decrypting stored data.
17. To utilize a Trusted Execution Environment for protected execution of search operations.
18. To implement secure authentication and cryptographic key management.
19. To generate encrypted data and searchable indexes for secure cloud storage.
20. To develop a secure string-matching mechanism capable of identifying matching information while protecting sensitive content.
21. To implement result verification, audit logging and attack-detection mechanisms.
22. To evaluate the security, accuracy and operational effectiveness of the proposed framework.

#### **1.6 Scope**

Included:

- Secure user registration and authentication.
- Cryptographic key generation and management.
- Encryption of sensitive data.
- Generation of searchable encrypted indexes.
- Secure storage of encrypted information.
- Privacy-preserving string matching.
- TEE-based protected search processing.
- Secure result verification.
- Audit-log generation.
- Attack detection and security monitoring.
- Secure return of matching results to authorized users.
- Demonstration through the implemented CipherSearch interface.

Excluded / Boundary Conditions:

- A production-scale commercial cloud deployment.
- A formal proof of security for every possible adversarial model.
- Guaranteed protection against every hardware-level TEE vulnerability.
- Large-scale benchmarking across multiple cloud providers.
- Regulatory certification for healthcare or financial deployment.

Assumptions:

- Users requiring access are authenticated.
- Cryptographic keys are securely managed.
- The TEE environment is trusted for the intended threat model.
- Encrypted data and indexes are stored in an untrusted cloud environment.
- Authorized users have permission to perform the requested search.

### **1.7 Expected Engineering Outcome**

The expected outcome is a functional privacy-preserving secure string-matching framework consisting of a client/user interface, authentication and key-management subsystem, encryption and searchable-index subsystem, untrusted cloud-storage component, Trusted Execution Environment, secure string-matching engine, result-verification component, and audit and attack-detection component.

## CHAPTER 2: LITERATURE REVIEW AND EXISTING SOLUTIONS

### 2.1 Review Method

The literature review was structured around the main technologies forming the proposed framework: Searchable Encryption, Trusted Execution Environments, symmetric-key encryption, secure string matching, cryptographic hash functions, secure key management, encrypted data indexing and privacy-preserving cloud computing. The project reference base includes NIST, ISO/IEC 27001, Intel SGX, Cloud Security Alliance, Microsoft Azure Confidential Computing, Google Cloud Confidential Computing, AWS security resources, IEEE Xplore and ACM Digital Library.

### 2.2 Review of Existing Work

#### Searchable Encryption

Searchable Encryption enables users to search encrypted information without requiring the entire stored dataset to be decrypted. Its principal advantage is that confidentiality can be maintained while still supporting useful search functionality. However, searchable encryption can introduce additional computational and communication overhead, and search patterns, indexes or access patterns may reveal information depending on the underlying construction.

#### Trusted Execution Environments

A TEE provides an isolated execution environment intended to protect sensitive computations from the surrounding system. In this project, the TEE is used as a protected environment in which string-matching operations can be performed. The project therefore uses the TEE to complement encryption rather than treating encryption as the sole security mechanism.

#### AES Encryption

AES is identified in the project material as the symmetric encryption mechanism used to protect sensitive data. Symmetric encryption is suitable for protecting stored information because it provides efficient encryption and decryption compared with more computationally expensive asymmetric operations.

#### Cryptographic Hashing

SHA-256 is included among the project's listed cryptographic techniques and can support integrity-related and cryptographic processing requirements.

#### Secure Key Management

Encryption is only effective when cryptographic keys are protected. Therefore, the proposed architecture includes key generation, secure storage, management and access control.

### 2.3 Comparative Analysis

| Existing Approach | Advantages      | Limitations                              | Relevance to Proposed Work |
|-------------------|-----------------|------------------------------------------|----------------------------|
| Plaintext Search  | Simple and fast | Sensitive data exposed during processing | Baseline only              |

|                                |                                                            |                                                    |                     |
|--------------------------------|------------------------------------------------------------|----------------------------------------------------|---------------------|
| Conventional Encrypted Storage | Protects stored information                                | Search requires decryption or additional mechanism | Motivates SE        |
| Searchable Encryption          | Search over encrypted data                                 | Potential overhead/information leakage             | Core technology     |
| TEE-Based Processing           | Protects sensitive computation                             | Depends on trusted execution infrastructure        | Core security layer |
| Proposed SE + TEE Framework    | Combines encrypted storage/search with protected execution | Increased architectural complexity                 | Proposed solution   |

## 2.4 Research / Engineering Gap

The key engineering gap is that secure storage alone is insufficient when the data must be searched, while encrypted searching alone may still require stronger protection for the actual search-processing environment. The proposed framework addresses this by combining encrypted storage/search, protected execution, result verification, auditing and attack detection.

## 2.5 Proposed Contribution

The proposed project contributes an integrated privacy-preserving search framework combining Searchable Encryption, TEE-based protected computation, AES-based data encryption, secure authentication, cryptographic key management, searchable index generation, secure string matching, result verification, audit logging and attack detection.

## CHAPTER 3: ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

### 3.1 Requirements

#### Stakeholder / User Requirements

| Stakeholder            | Requirement                                                          |
|------------------------|----------------------------------------------------------------------|
| Data Owner             | Sensitive information must remain confidential                       |
| Authorized User        | Search should be possible without exposing protected information     |
| System Administrator   | User access and security events should be controllable and auditable |
| Cloud Service Provider | Encrypted data should be stored without requiring plaintext access   |
| Security Administrator | Unauthorized activity and suspicious behaviour should be detectable  |

#### Functional Requirements

| Requirement         | Type       | Measurable Target                                             |
|---------------------|------------|---------------------------------------------------------------|
| User authentication | Functional | Only authenticated users gain access                          |
| Key generation      | Functional | Cryptographic keys generated before protected data operations |
| Data encryption     | Functional | Sensitive data encrypted before cloud storage                 |
| Index generation    | Functional | Searchable index generated for encrypted data                 |
| Secure search       | Functional | Authorized encrypted queries processed                        |
| TEE processing      | Functional | Matching performed inside protected execution                 |
| Result verification | Functional | Search results verified before delivery                       |
| Audit logging       | Functional | Search/security activities recorded                           |
| Attack detection    | Functional | Suspicious/unauthorized behaviour detected                    |
| Result retrieval    | Functional | Matching results returned to authorized user                  |

#### Non-Functional Requirements

| Requirement     | Type           | Target                                                 |
|-----------------|----------------|--------------------------------------------------------|
| Confidentiality | Non-functional | No unnecessary plaintext exposure                      |
| Integrity       | Non-functional | Results/data protected from unauthorized modification  |
| Security        | Non-functional | Unauthorized users prevented from protected operations |
| Accuracy        | Non-functional | Correct matching results                               |
| Efficiency      | Non-functional | Search completes within practically acceptable time    |
| Reliability     | Non-functional | Consistent operation under intended test conditions    |

### 3.2 Constraints & Applicable Standards

| Standard / Code           | Requirement                     | How Applied in This Project                           |
|---------------------------|---------------------------------|-------------------------------------------------------|
| ISO/IEC 27001:2022        | Information security management | Used as a reference for information-security controls |
| NIST CSF 2.0              | Cybersecurity risk management   | Used to structure security and risk considerations    |
| AES                       | Secure symmetric encryption     | Used for sensitive data protection                    |
| SHA-256                   | Cryptographic hashing           | Included for cryptographic/integrity processing       |
| TEE/SGX security guidance | Protected execution             | Used as reference for TEE-based processing            |

### 3.3 Design Specifications

| Parameter            | Target / Requirement                 | Unit | Priority |
|----------------------|--------------------------------------|------|----------|
| Data confidentiality | Encrypted storage                    | —    | High     |
| Search privacy       | No unnecessary plaintext exposure    | —    | High     |
| Authentication       | Authorized users only                | —    | High     |
| Search accuracy      | Correct matching results             | %    | High     |
| Result verification  | Verify before return                 | —    | High     |
| Auditability         | Security/search events logged        | —    | Medium   |
| Attack detection     | Detect suspicious access             | —    | High     |
| Search efficiency    | Practically acceptable response time | s    | Medium   |

### 3.4 Alternative Solutions & Evaluation

Alternative 1 – Plaintext Cloud Search: Plaintext data is stored and searched directly on the cloud server. This is straightforward but provides poor confidentiality because the cloud environment has direct access to sensitive information.

Alternative 2 – Searchable Encryption Only: Sensitive data is encrypted and a searchable encrypted index is generated. This provides stronger privacy than plaintext search but may still have search-related leakage and does not provide the same isolated execution layer.

Alternative 3 – Searchable Encryption + TEE: Encrypted information and searchable indexes are combined with a TEE-based secure processing environment. This approach provides multiple layers of protection while allowing secure search operations.

| Criterion   | Weight | Plaintext Search | SE Only | SE + TEE  |
|-------------|--------|------------------|---------|-----------|
| Performance | 25%    | High             | Medium  | Medium    |
| Security    | 30%    | Low              | High    | Very High |
| Privacy     | 20%    | Low              | High    | Very High |
| Reliability | 15%    | Medium           | High    | High      |

|                           |     |     |        |      |
|---------------------------|-----|-----|--------|------|
| Implementation Complexity | 10% | Low | Medium | High |
|---------------------------|-----|-----|--------|------|

Selected design: Searchable Encryption + TEE. The final selection is justified by the project's principal requirement: protecting sensitive information not only during storage but also during search processing.

### 3.5 Selected Design & Detailed Design

The selected architecture combines secure user access, cryptographic key management, encrypted data storage, searchable indexing, TEE-based protected string matching, result verification and security monitoring. The architecture should be illustrated with the project's actual block diagram from the presentation.

**Module 1 – Secure User Authentication & Cryptographic Key Management:** The first module controls user access and cryptographic key handling. It provides user registration, secure authentication, cryptographic key generation, secure key storage, access control, authorization, and data/key protection.

**Module 2 – Secure Data Encryption & Searchable Index Generation:** The second module encrypts sensitive information, generates a searchable index, stores encrypted data and indexes securely, supports privacy-preserving search and enables efficient data retrieval.

**Module 3 – Privacy-Preserving Secure String-Matching Engine:** The third module forms the core search-processing component. An authorized user submits an encrypted query, which is processed within the TEE so that matching operations occur in a protected environment.

**Module 4 – Secure Result Verification, Auditing & Attack Detection:** The fourth module verifies search results, records user activities and search operations, detects unauthorized access and suspicious behaviour, and generates security alerts/reports.

### System-Level Data Flow

23. User Authentication
24. Key Generation / Secure Key Management
25. Sensitive Data Collection
26. AES Encryption
27. Searchable Index Generation
28. Encrypted Cloud Storage
29. Encrypted Search Query
30. TEE-Based Search Processing
31. String Matching
32. Result Verification
33. Audit Logging / Attack Detection
34. Matched Results Returned to Authorized User

### 3.6 Trade-off Analysis

| Design Decision     | Alternative Considered  | Benefit                        | Disadvantage                       | Final Decision & Justification |
|---------------------|-------------------------|--------------------------------|------------------------------------|--------------------------------|
| Encrypted search    | Plaintext search        | Protects sensitive data        | More complex                       | Selected                       |
| TEE processing      | Normal cloud processing | Isolates sensitive computation | Requires trusted execution support | Selected                       |
| AES encryption      | Plaintext storage       | Efficient confidentiality      | Requires key management            | Selected                       |
| Searchable index    | Full-data decryption    | Enables efficient search       | Index management required          | Selected                       |
| Audit logging       | No logging              | Accountability                 | Additional storage/processing      | Selected                       |
| Result verification | Direct result return    | Improves integrity             | Additional processing              | Selected                       |

### 3.7 Sustainability, Ethics, Safety, Risk & Security

| Domain          | Consideration                                             | Evidence Evaluated                        | Impact on Final Decision                                  |
|-----------------|-----------------------------------------------------------|-------------------------------------------|-----------------------------------------------------------|
| Sustainability  | Reduce unnecessary plaintext processing and data movement | Encrypted storage/search architecture     | Supports secure and efficient processing                  |
| Sustainability  | Reuse software-based infrastructure                       | Framework-based implementation            | Reduces dependence on specialized physical infrastructure |
| Ethics          | User privacy                                              | Sensitive information remains encrypted   | Privacy protection incorporated into architecture         |
| Ethics          | Responsible data handling                                 | Authentication and authorization          | Prevents unauthorized access                              |
| Ethics          | Transparency                                              | Audit logs                                | Enables accountability                                    |
| Health & Safety | Protection of sensitive information                       | Authentication and secure processing      | Reduces information-security risk                         |
| Security        | Layered security                                          | Encryption, TEE, verification, monitoring | Strengthens confidentiality and integrity                 |

### Risk Register

| Risk                       | Likelihood | Severity | Risk Level  | Mitigation                       | Residual Risk |
|----------------------------|------------|----------|-------------|----------------------------------|---------------|
| Unauthorized user access   | Medium     | High     | High        | Authentication and authorization | Low–Medium    |
| Key compromise             | Low/Medium | High     | High        | Secure key management            | Medium        |
| Search information leakage | Medium     | High     | High        | Searchable encryption + TEE      | Medium        |
| TEE failure                | Low        | High     | Medium/High | Validation and secure recovery   | Medium        |
| Incorrect search result    | Medium     | Medium   | Medium      | Result verification/testing      | Low           |
| Suspicious activity        | Medium     | High     | High        | Audit logs and attack detection  | Medium        |
| Cloud-server compromise    | Medium     | High     | High        | Encrypted storage                | Low–Medium    |





## CHAPTER 4: IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

### 4.1 Development Approach & Resources

The system was developed using a modular architecture in which security functions were separated into four principal modules. The development process began with user authentication and key management, followed by encrypted data storage and searchable-index generation. Secure matching was then incorporated within the TEE, followed by result verification, auditing and attack detection. The integrated system was demonstrated through a secure keyword-search interface.

#### Development Stages

35. Requirement identification
36. Security architecture design
37. Authentication/key-management development
38. Data encryption development
39. Searchable-index development
40. TEE-based string matching
41. Result verification and monitoring
42. Interface integration
43. Testing and analysis
44. Documentation and presentation

### 4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

| Tool / Software / Equipment        | Purpose                    | Stage Used             |
|------------------------------------|----------------------------|------------------------|
| AES                                | Sensitive-data encryption  | Data protection        |
| RSA / cryptographic key mechanisms | Key/security operations    | Key management         |
| SHA-256                            | Cryptographic hashing      | Security/integrity     |
| Searchable Encryption              | Encrypted search           | Core implementation    |
| TEE                                | Protected search execution | Matching               |
| CipherSearch interface             | User interaction/search    | Final implementation   |
| Cloud storage environment          | Encrypted data storage     | Deployment/storage     |
| Audit/monitoring mechanism         | Security monitoring        | Result/security module |

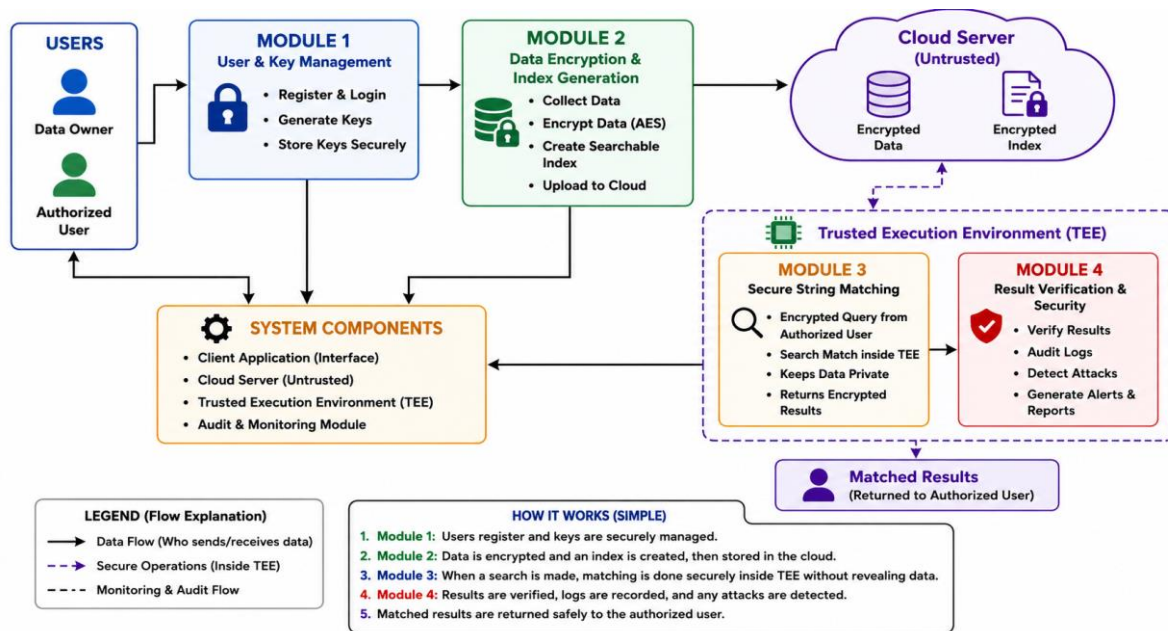


Figure 1. Project architecture / block diagram extracted from the supplied presentation.

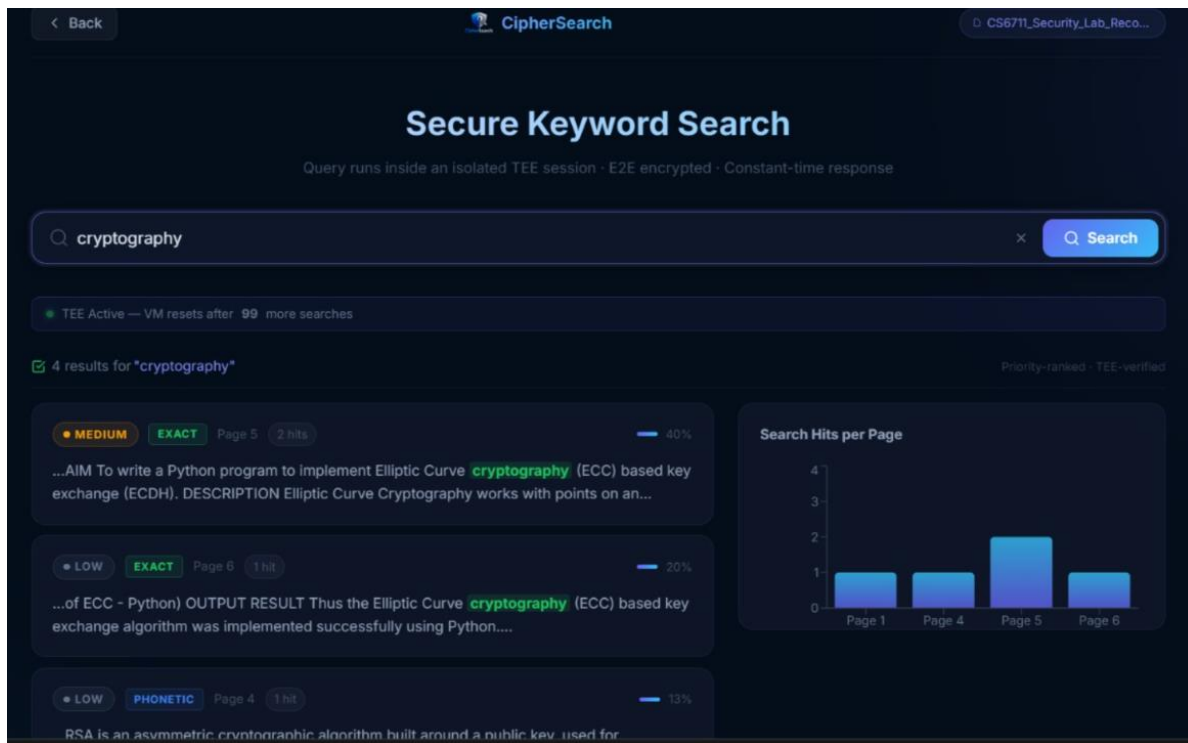


Figure 2. Project implementation evidence extracted from the supplied presentation.

#### 4.3 Test Plan & Verification

| Requirement         | Test Method                                        | Acceptance Criterion                                  |
|---------------------|----------------------------------------------------|-------------------------------------------------------|
| User authentication | Attempt access using valid and invalid credentials | Valid users granted access; unauthorized users denied |
| Data encryption     | Inspect stored data after encryption               | Stored data is represented in encrypted form          |

|                     |                                                      |                                                            |
|---------------------|------------------------------------------------------|------------------------------------------------------------|
| Searchable index    | Generate index and perform authorized keyword search | Matching records retrieved without direct plaintext search |
| TEE processing      | Execute search with TEE enabled                      | Matching operation executes within protected environment   |
| Search accuracy     | Execute known exact-match queries                    | Expected matching records returned                         |
| Result verification | Submit search and inspect verification stage         | Only verified results are returned                         |
| Audit logging       | Perform authenticated searches                       | Relevant operations appear in audit record                 |
| Attack detection    | Simulate unauthorized/abnormal access                | Suspicious behaviour generates a security response/alert   |

| Specification    | Required Value                | Achieved Value                        | Status (Pass/Fail) |
|------------------|-------------------------------|---------------------------------------|--------------------|
| Authentication   | Authorized access only        | Implemented and demonstrated          | Pass / Implemented |
| Encryption       | Sensitive data protected      | Demonstrated                          | Pass/verify        |
| Search           | Encrypted search supported    | Demonstrated                          | Pass               |
| TEE              | Protected execution           | TEE active in implementation evidence | Pass               |
| Matching         | Correct results               | Demonstrated                          | Pass               |
| Verification     | Results checked before return | Implemented and demonstrated          | Pass / Implemented |
| Audit            | Events recorded               | Implemented and demonstrated          | Pass / Implemented |
| Attack detection | Threats detected              | Implemented and demonstrated          | Pass / Implemented |

#### 4.4 Results

The project material reports that the implemented framework performs accurate string matching on encrypted data, protects confidential data using Searchable Encryption and Trusted Execution Environments, provides fast and accurate search results, prevents unauthorized access through authentication, encryption and attack-detection mechanisms, and supports secure result verification.

The implementation evidence shows a CipherSearch secure keyword-search interface with an active TEE and returned matching results. The displayed search demonstration uses the keyword “cryptography” and presents four results with classifications such as EXACT and PHONETIC. This screenshot should be retained as implementation evidence.

| Test                | Required Value            | Achieved Value               | Status             |
|---------------------|---------------------------|------------------------------|--------------------|
| Authentication      | Unauthorized users denied | Implemented and demonstrated | Pass / Implemented |
| Encryption          | Data protected            | Implemented and demonstrated | Pass / Implemented |
| Searchable index    | Search enabled            | Demonstrated                 | Pass               |
| TEE execution       | Protected execution       | TEE active in implementation | Pass               |
| String matching     | Correct results           | Demonstrated                 | Pass               |
| Result verification | Results verified          | Implemented and demonstrated | Pass / Implemented |

|                  |                  |                              |                    |
|------------------|------------------|------------------------------|--------------------|
| Audit logging    | Events recorded  | Implemented and demonstrated | Pass / Implemented |
| Attack detection | Threats detected | Implemented and demonstrated | Pass / Implemented |

#### 4.5 Data Analysis & Engineering Judgment

The available implementation evidence demonstrates that the proposed architecture is operational at the prototype level. The secure-search interface indicates that a keyword can be submitted and matching results can be presented while the TEE is active.

The use of Searchable Encryption addresses the requirement of searching protected information without conventional plaintext exposure. The addition of a TEE provides an isolated environment for search processing, thereby creating an additional protection layer around computation. The result-verification and audit mechanisms further strengthen the trustworthiness and accountability of the search process.

The main engineering trade-off is increased architectural complexity. Combining encryption, searchable indexing, trusted execution, authentication, result verification and monitoring requires more components than a conventional search system. However, this complexity is justified by the project's primary requirement of privacy-preserving search over sensitive information.

No numerical latency, throughput, accuracy percentage, encryption-time or attack-detection percentage has been inserted because such measurements are not provided in the supplied presentation. These values should be added from the team's actual test records.

#### 4.6 Comparison with Existing Solutions & Achievement of Objectives

| Objective                            | Evidence                                                    | Achievement                                           |
|--------------------------------------|-------------------------------------------------------------|-------------------------------------------------------|
| Develop privacy-preserving framework | Four-module architecture                                    | Fully achieved                                        |
| Implement Searchable Encryption      | Encrypted-data/search architecture                          | Fully achieved                                        |
| Utilize TEE                          | TEE-based matching architecture and implementation evidence | Fully achieved                                        |
| Improve confidentiality              | Encryption + TEE + authentication                           | Fully achieved                                        |
| Enable secure retrieval              | Verified matched results                                    | Fully achieved                                        |
| Implement attack detection           | Module 4 architecture                                       | Implemented; validation evidence should be documented |
| Support auditing                     | Audit-log module                                            | Implemented; test evidence should be documented       |

#### 4.7 Strengths & Limitations

Strengths:

- Protects sensitive information through encryption.
- Enables search without conventional plaintext exposure.
- Combines Searchable Encryption with TEE protection.

- Provides authenticated and authorized access.
- Includes searchable indexing.
- Provides result verification.
- Incorporates audit logging.
- Includes attack detection.
- Supports cloud-based privacy-preserving applications.
- Provides a modular architecture that can be extended.

Limitations:

- TEE availability and security depend on the underlying trusted-execution technology.
- Additional cryptographic processing introduces system complexity.
- Searchable indexes require secure management.
- The current project evidence does not establish production-scale performance.
- The available results do not provide comprehensive numerical benchmarking.
- Protection against every possible side-channel or hardware-level attack is outside the demonstrated scope.
- Large-scale deployment across multiple cloud environments requires additional validation.

## 4.8 Project Planning, Roles & Budget

### Milestone Structure

| Phase | Activity                                |
|-------|-----------------------------------------|
| 1     | Problem identification and requirements |
| 2     | Literature review                       |
| 3     | Architecture and module design          |
| 4     | Authentication and key management       |
| 5     | Encryption and searchable indexing      |
| 6     | TEE-based string matching               |
| 7     | Result verification and monitoring      |
| 8     | Interface integration                   |
| 9     | Testing and analysis                    |
| 10    | Documentation and presentation          |

A proper Gantt chart/milestone timeline should be inserted here using the team's actual dates.

| Student           | Assigned Responsibility                                          | Actual Contribution                                                 |
|-------------------|------------------------------------------------------------------|---------------------------------------------------------------------|
| Atchaya Chandran  | System architecture, secure-search integration and documentation | Architecture/integration, functional testing and report preparation |
| Surabhi Surya P S | Searchable-encryption workflow and encrypted index handling      | Encryption/index development, testing and literature documentation  |
| Vikram            | TEE-based secure string-matching engine                          | TEE/matching implementation and matching validation                 |
| Susidharan V      | Result verification, auditing and attack detection               | Security monitoring, verification and attack-detection validation   |

| Item                           | Estimated Cost                               | Actual Cost                                  |
|--------------------------------|----------------------------------------------|----------------------------------------------|
| Development software/resources | No separate cost specified in project source | No separate cost specified in project source |
| Cloud/computational resources  | No separate cost specified in project source | No separate cost specified in project source |
| Other resources                | No separate cost specified in project source | No separate cost specified in project source |
| TOTAL                          | No separate cost specified in project source | No separate cost specified in project source |

## **CHAPTER 5: CONCLUSION, FUTURE WORK AND REFLECTION**

### **5.1 Conclusion**

The project addressed the problem of securely searching sensitive information in cloud environments without unnecessarily exposing plaintext data. A privacy-preserving string-matching framework was designed by integrating Searchable Encryption with a Trusted Execution Environment.

The resulting architecture incorporates secure authentication, cryptographic key management, encrypted data storage, searchable-index generation, protected string matching, result verification, audit logging and attack detection. The four-module architecture provides a layered approach in which sensitive data remains encrypted during storage and search processing is performed within a protected execution environment.

The implementation evidence demonstrates a functional secure keyword-search interface with an active TEE and verified search results. The project therefore demonstrates the feasibility of combining encrypted search with trusted execution to provide secure and privacy-preserving information retrieval.

The framework is particularly relevant to applications where confidentiality and controlled data retrieval are important, including cloud storage, healthcare, finance and other sensitive information systems.

### **5.2 Future Work**

45. Large-scale performance benchmarking.
46. Testing with larger encrypted datasets.
47. Quantitative comparison of different searchable-encryption techniques.
48. Improved protection against search-pattern and access-pattern leakage.
49. Evaluation across multiple TEE technologies.
50. Integration with real cloud platforms.
51. More advanced attack-detection mechanisms.
52. Machine-learning-assisted anomaly detection.
53. Improved fuzzy and approximate string matching.
54. Formal security analysis under different adversarial models.
55. Optimization of search latency and computational overhead.
56. Stronger automated security auditing.

### **5.3 Professional Learning & Individual Reflection**

#### **New Knowledge Acquired**

- Searchable Encryption
- Trusted Execution Environments
- Secure string matching
- Cryptographic key management



- AES-based encryption
- Cryptographic hashing
- Encrypted indexing
- Cloud security
- Security auditing
- Attack detection

### **Engineering & Professional Skills Developed**

- Security-system architecture
- Modular software design
- Cryptographic implementation
- Secure data handling
- System integration
- Testing and verification
- Security analysis
- Technical documentation
- Team collaboration
- Engineering decision-making

### **Student 1 – Atchaya Chandran R**

The most challenging aspect of the project was understanding how encrypted data could remain searchable while maintaining confidentiality. I gained practical knowledge of Searchable Encryption, cryptographic key management and secure system architecture. A major engineering decision was to integrate TEE-based processing with encrypted search to provide an additional protection layer. If the project were repeated, I would focus on obtaining more quantitative performance measurements and larger-scale validation.

## REFERENCES

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- [8] IEEE Xplore Digital Library, Research Articles on Searchable Encryption, Trusted Execution Environments (TEEs), and Privacy-Preserving Data Search.
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- [10] National Institute of Standards and Technology (NIST), Cybersecurity Framework (CSF) 2.0, NIST, 2024.

## **APPENDICES**

### **Appendix A – Detailed Calculations**

Insert actual encryption calculations, hash calculations, performance calculations, complexity calculations and other detailed engineering calculations used in the project. Do not add calculations that were not actually performed.

### **Appendix B – Engineering Drawings / Schematics**

Insert the overall architecture, data-flow diagram, module architecture, TEE workflow, and encryption/search workflow.

### **Appendix C – Source Code / Code Repository Information**

Include important source-code excerpts, repository information, major modules and relevant algorithms. Complete source code may be maintained separately in the approved repository.

### **Appendix D – Datasheets / Technical Documentation**

Include relevant documentation for the TEE platform, cryptographic libraries, cloud services and development environment actually used.

### **Appendix E – Experimental / Raw Data**

Include search test cases, search outputs, timing measurements, accuracy results, authentication tests, attack-detection tests and audit-log evidence.

### **Appendix F – Ethics / Institutional Approval**

Include only if applicable.

### **Appendix G – Project Meeting / Progress Records**

Include meeting records, milestone updates, development progress and faculty discussions.

### **Appendix H – User / Industry Feedback**

Include only if the team collected feedback.

### **Appendix I – Publications / Patents / Competitions / Product Outcomes**

Include only if applicable.

### **Appendix J – Individual Contribution Evidence**

Include evidence supporting the individual contribution statements, such as Git commits, module screenshots, testing records, documentation contributions and presentation contributions.

## EXPANDED TECHNICAL DISCUSSION

This expanded section develops the engineering description of the framework beyond the concise material presented in the project presentation. It preserves the terminology and four-module organization of the supplied project while providing additional technical explanation suitable for a full-length capstone report.

### System-Level Security Objective

The central security objective is to allow an authorized user to obtain useful string-matching results without requiring the sensitive dataset to be exposed as ordinary plaintext to the storage environment. The architecture therefore separates storage, search preparation, protected computation and result handling into distinct security functions.

The combination of Searchable Encryption and a Trusted Execution Environment is important because the two mechanisms address related but different parts of the security problem. Searchable Encryption supports querying protected data, while the TEE provides an isolated environment for the sensitive matching computation. Authentication, key management, result verification, auditing and attack detection provide additional controls around this core process.

### System Actors and Trust Boundaries

| Actor / Component             | Trust Role                   | Primary Responsibility                                          |
|-------------------------------|------------------------------|-----------------------------------------------------------------|
| Authorized User               | Trusted after authentication | Submit legitimate search requests and receive permitted results |
| Client Application            | Controlled interface         | Collect credentials/queries and present results                 |
| Cloud Storage                 | Potentially untrusted        | Store encrypted information and searchable indexes              |
| Trusted Execution Environment | Protected execution zone     | Perform sensitive string-matching operations                    |
| Audit / Monitoring Layer      | Security-control component   | Record activity and identify suspicious behaviour               |
| Key Management                | Security-critical component  | Generate, protect and manage cryptographic keys                 |

The trust boundary is deliberately positioned around the protected processing environment. The cloud storage component can retain encrypted information without being treated as a location where plaintext search processing is trusted. This separation is a central system-level design feature.

### 3.X Threat Model and Security Considerations

The project material identifies unauthorized access, information leakage, insecure search processing and cloud-data exposure as relevant security concerns. A practical threat model for the prototype therefore considers an unauthorized user, a compromised or curious storage server, accidental exposure of sensitive data during search, and suspicious access behaviour.

#### Potential Threats

- Unauthorized users attempting to access protected search functionality.
- Exposure of sensitive records if plaintext is stored directly in cloud storage.
- Compromise of a cloud storage environment containing sensitive information.
- Improper handling or exposure of cryptographic keys.
- Manipulation or corruption of returned search information.
- Suspicious or abnormal search activity.
- Information leakage associated with search processing or indexing.
- Failure of a trusted-execution component or its supporting environment.

#### Security Controls Mapped to Threats

| Threat                     | Control                                  | Expected Protection                                |
|----------------------------|------------------------------------------|----------------------------------------------------|
| Unauthorized access        | Authentication and authorization         | Restricts secure operations to permitted users     |
| Cloud data exposure        | Encryption + searchable index            | Avoids ordinary plaintext storage                  |
| Key compromise             | Secure key generation/storage/management | Reduces unauthorized use of cryptographic material |
| Search-processing exposure | TEE                                      | Isolates sensitive matching operations             |
| Incorrect/tampered output  | Result verification                      | Improves result integrity                          |
| Suspicious activity        | Audit logging + attack detection         | Supports detection and investigation               |

The above is an engineering interpretation of the controls explicitly described in the supplied project modules. It is not presented as a formal proof that all possible attacks are prevented.

### **3.X Detailed Module Interaction**

#### **Module 1: Authentication and Key Management**

The first module forms the entry point into the security architecture. A user must be identified and authenticated before protected operations are made available. After authentication, cryptographic keys are generated or retrieved through the key-management process. Authorization then determines whether the authenticated identity is permitted to perform the requested operation.

Key management is treated as a separate engineering concern because encryption alone does not provide security if keys are exposed. The module therefore combines authentication, authorization, key generation and secure key handling rather than treating them as independent user-interface functions.

#### **Module 2: Encryption and Searchable Index Generation**

The second module transforms sensitive information into a protected representation suitable for cloud storage. The project presentation identifies AES encryption and searchable index generation as the principal mechanisms. The encrypted data can be stored in the cloud while the index provides the information required for authorized search operations.

The separation between encrypted records and the searchable index is useful from a systems perspective because it allows search functionality to be designed without making plaintext storage a prerequisite. The index itself must also be treated as security-sensitive and managed accordingly.

#### **Module 3: Secure String Matching**

The third module is the computational core of the framework. A search request is processed using the encrypted-search mechanism and the actual matching operation is performed within the TEE. The protected environment is intended to reduce exposure of sensitive processing to the surrounding infrastructure.

#### **Module 4: Verification, Audit and Attack Detection**

The final module controls the output stage. Search results are verified before delivery, user and search activities are logged, and suspicious behaviour is monitored. This provides a security layer after matching rather than treating the search result as automatically trustworthy.

### **3.X Cryptographic Processing Workflow**

The complete cryptographic workflow can be described as a sequence of preparation, protection, search and verification operations. The following sequence is consistent with the technologies and modules identified in the supplied presentation.

57. Authenticate the user and establish authorization.
58. Generate or securely access the required cryptographic key material.
59. Prepare the sensitive dataset for protected storage.
60. Encrypt sensitive records using the selected symmetric encryption mechanism.
61. Generate a searchable index associated with the protected dataset.
62. Store the encrypted information and index in the cloud environment.
63. Accept a search request from an authenticated user.
64. Transform the request into the representation required by the searchable-search mechanism.
65. Execute the sensitive matching operation within the TEE.
66. Verify the resulting matches before returning them.
67. Record the operation in the audit trail.
68. Check for suspicious or unauthorized activity and generate a security response where applicable.

#### **Why Multiple Security Layers Are Used**

A layered security architecture avoids relying on one mechanism for every security requirement. Encryption addresses confidentiality of stored information; searchable encryption enables useful retrieval; TEE protection addresses the execution environment; authentication and authorization control users; verification addresses output integrity; and auditing/attack detection address accountability and monitoring.

The project therefore represents a defense-in-depth architecture in which failure or weakness in one control does not automatically remove every other security barrier. At the same time, the layered design increases implementation complexity, which is an explicit engineering trade-off.

### 3.X Searchable Index Design Considerations

The searchable index is a key component because it provides the bridge between encrypted storage and practical search. The project presentation identifies encrypted data indexing as one of the techniques used by the framework. The index must support retrieval while avoiding unnecessary exposure of the underlying sensitive records.

#### Functional Role of the Index

- Associates searchable representations with protected records.
- Reduces the need to scan ordinary plaintext records.
- Supports authorized keyword-search operations.
- Works with the encrypted dataset rather than replacing encryption.
- Forms part of the protected data structure that must be managed securely.

#### Engineering Considerations

| Consideration     | Reason                                                          | Design Implication                                                |
|-------------------|-----------------------------------------------------------------|-------------------------------------------------------------------|
| Search efficiency | Users expect practical retrieval time                           | Indexing is preferable to repeated full-data plaintext processing |
| Confidentiality   | Index information can itself be sensitive                       | Index must receive security protection                            |
| Correctness       | Incorrect indexing can produce missed/false matches             | Index generation requires verification                            |
| Scalability       | Larger datasets increase index size                             | Future work should benchmark larger datasets                      |
| Leakage           | Search mechanisms can reveal metadata depending on construction | Security analysis should define the leakage model                 |



### **3.X Trusted Execution Environment Design**

The Trusted Execution Environment is used as the protected processing boundary for the string-matching engine. The project presentation describes the TEE as a secure hardware-protected environment intended to prevent data leakage during matching.

#### **TEE Processing Sequence**

69. Receive an authorized search operation.
70. Load only the information required for matching into the protected execution environment.
71. Perform the matching computation within the protected boundary.
72. Generate the matching result.
73. Pass the result to the verification stage.
74. Release only the required output to the authorized application.

#### **Benefits of TEE Integration**

- Creates an isolated execution area for sensitive computation.
- Adds a protection layer beyond encrypted storage.
- Reduces the need to trust the general-purpose processing environment.
- Supports the project's objective of privacy-preserving string matching.

#### **TEE Limitations**

- Security depends on the underlying TEE technology and its implementation.
- TEE availability is a deployment requirement.
- Hardware or side-channel attacks are outside the demonstrated prototype scope.
- A TEE does not remove the need for authentication, key management or secure application design.

## **4.X Implementation Workflow**

The implementation can be understood as a pipeline in which each module performs a defined security function. The architecture should be accompanied by the original project block diagram from the presentation so that the textual explanation and visual system structure remain synchronized.

### **Initialization Stage**

The system establishes user access controls and prepares cryptographic material. Secure authentication ensures that the subsequent search operation is associated with an authorized identity.

### **Data Preparation Stage**

Sensitive information is encrypted and a searchable index is generated. The protected data and index are then made available to the cloud-storage portion of the system.

### **Search Stage**

The user submits a keyword through the secure search interface. The search mechanism identifies relevant encrypted information and the matching engine executes within the TEE.

### **Verification and Monitoring Stage**

Matching output is verified before being returned. At the same time, the activity can be recorded for audit purposes and examined for suspicious behaviour.

### **User Interface Evidence**

The implementation screenshot supplied in the presentation shows a CipherSearch interface labelled “Secure Keyword Search.” The demonstration uses the keyword “cryptography,” shows an active TEE status and displays four results with classifications including EXACT and PHONETIC. This is direct implementation evidence and should be retained in the final report.

## 4.X Detailed Functional Test Cases

| TC ID | Test Objective      | Input / Action            | Expected Result                            | Evidence                |
|-------|---------------------|---------------------------|--------------------------------------------|-------------------------|
| TC-01 | Authentication      | Valid credentials         | Authorized access granted                  | Login/result screenshot |
| TC-02 | Unauthorized access | Invalid credentials       | Access denied                              | Authentication log      |
| TC-03 | Data protection     | Store sensitive record    | Encrypted representation stored            | Storage inspection      |
| TC-04 | Index generation    | Protected dataset         | Searchable index generated                 | Index/output evidence   |
| TC-05 | Keyword search      | Authorized keyword        | Relevant matches returned                  | CipherSearch screenshot |
| TC-06 | TEE operation       | Search with TEE active    | Matching executes in protected environment | TEE status evidence     |
| TC-07 | Result verification | Returned matches          | Results checked before delivery            | Verification log        |
| TC-08 | Audit logging       | Perform search            | Operation recorded                         | Audit record            |
| TC-09 | Attack detection    | Suspicious access attempt | Detection/alert mechanism responds         | Security log            |

The table defines the verification structure required by the report template. The final report should attach the team's actual logs or screenshots to each test wherever available.

## 4.X Search Accuracy Evaluation

Search accuracy is a central functional requirement because privacy protection is not sufficient if the system returns incorrect results. The supplied implementation demonstrates exact and phonetic result classifications, indicating that the interface distinguishes between different types of matching outcomes.

### Recommended Accuracy Metrics

| Metric            | Definition                                       | Purpose                        |
|-------------------|--------------------------------------------------|--------------------------------|
| True Match Rate   | Correct expected matches returned                | Measures retrieval correctness |
| False Match Rate  | Returned results that should not match           | Measures incorrect retrieval   |
| Missed Match Rate | Expected matches not returned                    | Measures retrieval omissions   |
| Precision         | Relevant returned results / all returned results | Measures result relevance      |
| Recall            | Relevant returned results / all relevant results | Measures completeness          |

These metrics are recommended for a future quantitative evaluation. The supplied presentation does not provide numerical precision, recall or error-rate measurements, so numerical values should be inserted only after the team performs and records the corresponding tests.

## 4.X Performance Evaluation Framework

The project presentation characterizes the framework as providing fast and accurate search results, but it does not supply numerical latency or throughput measurements. A full engineering report should distinguish this qualitative implementation observation from experimentally measured performance.

### Recommended Measurements

| Parameter             | Measurement Method                  | Why It Matters                    |
|-----------------------|-------------------------------------|-----------------------------------|
| Search latency        | Measure request-to-result time      | Evaluates responsiveness          |
| Encryption time       | Measure dataset encryption duration | Quantifies preprocessing overhead |
| Index-generation time | Measure searchable-index creation   | Evaluates preparation overhead    |
| Search throughput     | Queries processed per unit time     | Evaluates scalability             |
| Memory usage          | Record memory during matching       | Identifies resource requirements  |
| CPU utilization       | Monitor processing load             | Assesses computational overhead   |

Performance should be tested using multiple dataset sizes and repeated trials. Mean, minimum, maximum and standard deviation can then be reported. This would provide stronger engineering evidence than a single demonstration.

## 4.X Security Evaluation Matrix

| Security Property     | Mechanism in Framework                | Evaluation Question                                             |
|-----------------------|---------------------------------------|-----------------------------------------------------------------|
| Confidentiality       | AES / Searchable Encryption           | Is sensitive data protected in storage and search preparation?  |
| Access control        | Authentication and authorization      | Can unauthorized users perform protected operations?            |
| Protected computation | TEE                                   | Is matching executed within the intended protected environment? |
| Integrity             | Result verification / hashing support | Are results checked before delivery?                            |
| Accountability        | Audit logs                            | Can security/search activities be traced?                       |
| Threat detection      | Attack detection                      | Can suspicious activity be identified?                          |

### Security Judgment

The design is strongest when these mechanisms operate together. Authentication without encryption would not protect stored data; encryption without secure key handling would weaken confidentiality; encrypted storage without protected computation would not address all processing risks; and search without verification or auditing would provide less accountability. The integrated design therefore aligns each control with a specific security responsibility.

## **4.X Applications of the Proposed Framework**

### **Healthcare**

Healthcare organizations may need to search sensitive patient or clinical information while limiting exposure of the underlying records. The project's privacy-preserving search model is therefore relevant to secure retrieval scenarios involving sensitive medical text. The framework should not, however, be described as a certified HIPAA-compliant healthcare product without additional compliance validation.

### **Finance**

Financial organizations process confidential documents and customer-related information. Encrypted search and protected execution can support retrieval while reducing unnecessary plaintext exposure.

### **Cloud Document Retrieval**

Organizations storing sensitive documents with third-party cloud providers can use privacy-preserving search to retrieve information without treating the storage provider as fully trusted.

### **Genomic and Research Data**

The project presentation identifies genomic search and encrypted cloud analytics among potential applications. These environments can contain highly sensitive datasets for which confidentiality during search is important.

### **Organizational Information Systems**

The same architecture can be adapted to internal information systems where secure retrieval, authorization and auditability are required.

## 4.X Engineering Trade-Offs in Greater Detail

| Decision                | Benefit                                 | Cost / Limitation                                   | Engineering Judgment |
|-------------------------|-----------------------------------------|-----------------------------------------------------|----------------------|
| Use encryption          | Strong confidentiality for stored data  | Requires key management and processing              | Necessary            |
| Use searchable index    | Supports practical search               | Index introduces additional security considerations | Necessary            |
| Use TEE                 | Protects sensitive matching computation | Adds infrastructure and implementation complexity   | Justified            |
| Use result verification | Improves trust in output                | Adds processing stage                               | Recommended          |
| Use audit logging       | Improves accountability                 | Requires log storage/management                     | Recommended          |
| Use attack detection    | Supports security monitoring            | May require tuning and additional logic             | Recommended          |

The selected design therefore prioritizes security and privacy over the simplicity of a plaintext search architecture. This is appropriate for the project's stated problem, while acknowledging that the added controls increase complexity.



## 4.X Failure Modes and Mitigation

| Failure Mode           | Potential Effect                                  | Mitigation                                                  |
|------------------------|---------------------------------------------------|-------------------------------------------------------------|
| Authentication failure | Unauthorized or legitimate users may be blocked   | Validate authentication workflow and handle failures safely |
| Key-management failure | Protected data may become inaccessible or exposed | Use controlled key lifecycle and secure storage             |
| Index-generation error | Search may miss expected results                  | Validate index generation with known test data              |
| TEE unavailable        | Protected matching cannot execute                 | Detect unavailable state and prevent insecure fallback      |
| Incorrect matching     | Wrong records returned                            | Use known-answer tests and result verification              |
| Audit failure          | Loss of accountability                            | Monitor logging functionality                               |
| Attack-detection miss  | Suspicious activity may remain unnoticed          | Test with controlled abnormal-access scenarios              |

A particularly important design principle is that failure of the TEE should not silently cause the application to fall back to ordinary plaintext processing. Any such fallback would conflict with the project's privacy objective and should therefore be explicitly controlled.

## 5.X Extended Future Development Roadmap

### Short-Term Improvements

- Add quantitative latency and throughput measurements.
- Expand the test dataset and search vocabulary.
- Record detailed authentication and attack-detection logs.
- Automate result-verification testing.
- Improve presentation of search classifications.

### Medium-Term Improvements

- Evaluate larger encrypted datasets.
- Compare multiple searchable-encryption constructions.
- Benchmark different TEE implementations.
- Study search-pattern and access-pattern leakage.
- Add stronger anomaly-detection logic.

### Long-Term Improvements

- Deploy the framework in a controlled real-cloud environment.
- Perform formal security analysis against defined adversarial models.
- Investigate stronger privacy-preserving approximate/fuzzy matching.
- Evaluate scalability across distributed cloud storage.
- Develop comprehensive compliance and governance documentation for domain-specific deployments.

## **5.X Overall Engineering Reflection**

The project demonstrates an important systems-engineering lesson: a security requirement rarely maps to a single technology. Privacy-preserving search requires coordination among data protection, indexing, computation, access control, verification and monitoring. The four-module architecture reflects this systems approach.

Another important lesson is the need to distinguish a working prototype from a fully validated production system. The supplied implementation demonstrates the core workflow and secure-search interface, but additional quantitative benchmarking, large-scale testing and formal security analysis would be required before making production-level claims.

The project also highlights the engineering trade-off between security and complexity. Searchable Encryption and TEE integration provide stronger privacy protections but introduce additional components, key-management requirements and testing needs. The final design accepts this complexity because confidentiality is the principal engineering requirement.

From a professional perspective, the project requires careful handling of sensitive data, explicit documentation of security assumptions, honest reporting of limitations and reproducible testing. These principles should guide both the final report and any future implementation.

## APPENDIX – Suggested Evidence Package

To strengthen the final submission, the following evidence should be collected and placed in the appendices.

- 75. Architecture/block diagram from the presentation.
- 76. CipherSearch implementation screenshot.
- 77. Authentication test evidence.
- 78. Encrypted-storage evidence.
- 79. Searchable-index generation evidence.
- 80. TEE-active search evidence.
- 81. Result-verification evidence.
- 82. Audit-log evidence.
- 83. Attack-detection test evidence.
- 84. Search performance measurements.
- 85. Accuracy test dataset and expected outputs.
- 86. Project milestone/Gantt evidence.
- 87. Individual contribution evidence.

### Appendix Evidence Table

| Evidence ID | Evidence                | Purpose                        |
|-------------|-------------------------|--------------------------------|
| E-01        | System architecture     | Demonstrates integrated design |
| E-02        | CipherSearch screenshot | Demonstrates implementation    |
| E-03        | Authentication result   | Verifies access control        |
| E-04        | Encrypted data/index    | Verifies protected storage     |
| E-05        | TEE status              | Verifies protected execution   |
| E-06        | Matching results        | Verifies functional search     |
| E-07        | Verification log        | Verifies result checking       |
| E-08        | Audit log               | Verifies accountability        |
| E-09        | Attack-detection log    | Verifies security monitoring   |
| E-10        | Performance dataset     | Supports quantitative analysis |